

Intel launches 3D Lakefield chips for foldable and ultra-thin laptops

Intel has launched new hybrid processors, which have been codenamed, Lakefield. These chipsets are made for next-gen PCs with foldable and dual-screen devices of the future.

The Intel Core i5 and i3 Lakefield processors with Intel hybrid technology are built for intense workloads and foreground applications while balancing power and performance optimisation for background tasks.

The Lakefield chipsets are completely compatible with Windows 10 application in up to a 56 per cent smaller

package area for up to 47 per cent smaller board size and extended battery life. This will enable original equipment manufacturers (OEMs) a more flexible form factor across single, dual, and foldable screen devices.

The company claims that these will be the first Intel Core processors to deliver as low as 2.5mW of standby system-on-chip (SoC) power—an up to 91 per cent reduction compared to Y-series processors for more time between charges.

AI-based system to track social distancing in public places

Researchers at IIT-Kharagpur have developed a cyber-physical system based on artificial intelligence for monitoring social distancing in public places. The team, under the leadership of professor Debashish Chakravarty and professor Aditya Bandyopadhyay, has engineered a low-cost device which can detect the gap between individuals, a spokesperson of IIT-KGP said.

“The device will play a proximity alert sound through audio output, whenever any violation of the social distancing norms occurs. It captures images of the field view and com-

putes the distance as per criteria set by the Ministry of Health and Family Welfare,” a member of the research group says.

The researchers have focused on the ease of deployment of such a device in remote places and curtailing the cost by using inexpensive and easily accessible hardware materials. Students have come up with the device leveraging readily-available hardware while the lockdown was in place. The system has been successfully tested by installing three prototypes of the device within the IIT-Kharagpur campus.

Now digitise your dog into a computer game

Researchers from the University of Bath have developed a motion capture technology that enables you to digitise your dog without a motion capture suit and using only one camera.

Motion capture technology is widely used in the entertainment industry, where actors wear a suit dotted with white markers, which are then precisely tracked in 3D space by multiple cameras taking images from different angles. Movement

data can then be transferred onto a digital character for use in films or computer games.

However, these technologies—particularly when applying them to animals—require expensive equipment and dozens of markers to be attached. Computer scientists at CAMERA at the University of Bath collected the movement data of a range of dogs to produce a model that could predict the poses of dogs from images taken by a single RGBD camera.



Researchers from the University of Bath have developed motion capture technology that can digitise your dog (Credit: University of Bath)

Wearing special doggie motion capture suits with markers, the dogs were filmed under the supervision of their BCDH handlers doing a range of movements as part of their enrichment activities.

They used this data to create a computer model that can accurately predict and replicate the poses of dogs when they're filmed without wearing the motion capture suits. This model allows 3D digital information for new dogs—their shape and movement—to be captured without markers and expensive equipment and just using a single RGBD camera. While normal digital cameras record the red, green and blue (RGB) colour in each pixel in the image, RGBD cameras also record the distance from the camera for each pixel.